

利用光栅衍射法测波长

衍射屏位置 70.00 cm

接收屏 3.00 cm

发射器 117.00 cm

序号 n	x_{n+1} (mm)	x_n (mm)	$x_n - x_{n-1}$ (mm)	λ (nm)
1	73.745	31.445	42.300	631.3
2	74.151	31 31.145	43.006	641.9
3	73.725	31.127	42.598	635.8
4	73.640	30 31.019	42.621	636.1
5	73.720	31.132	42.588	635.6
6	73.607	31.132 31.132 30.901	42.5 42.706	637.4

何铭源

3240104481

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李柏旭

2025.10.15

利用单缝衍射测狭缝宽度

序号 n	x_{n-1} (mm)	x_n (mm)	Δx (mm)	a (μm)
1	60.162 60.162 60.450 60.162	42.915	17.247	70.6
2	60 59.680	43.005 43.005	16.676	73.0
3	59.460	42.540	16.920	72.0
4	59.330	42.330	17.000	71.6
5	59.671	42.112	17.559	69.3
6	59.955	42.055	17.405	69.9

序号	全标 (mm)	光强	
1	38.030	9.3	
2	38.931	11.6	
3	39.300	7.6	
4	39.700	8.8	
5	40.100	57.0	
6	40.550	93.5	何锦源 3240104481
7	40.900	126.6	16
8	41.300	159.8	李相旭.
9	41.700	180.6	2025.10.15.
10	42.100	159.8 186.8	
11	42.500	174.2	
12	42.900	148.3	
13	43.300	107.7	
14	43.700	67.9	
15	44.100	30.4	
16	44.500	11.2	
17	44.900	28.2	
18	45.300	78.0	
19	45.700	192.4	
20	46.100	367.7	
21	46.500	602.0	
22	46.900	907.4	
23	47.300	1292.5	
24	47.700	1638.2	

序号	坐标 (mm)	光强
25	47 48.110	2057.8
26	48.500	2495.8
27	48.950	2884.7
28	49.300	3166.7
29	49.700	3438.5
30	50.100	3620.6
31	50.500	3698.0
32	50.900	3679.4
33	51.300	3557.4
34	51.700	3336.1
35	52.100	3042.3
36	52.500	2626.6
37	52.950	2207.5
38	53.300	1865.2
39	53.700	1450.6
40	54.100	1072.1
41	54.550	672.4
42	54.900	468.9
43	55.300	267.1
44	55.700	127.3

何铭源

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李振旭.

2025.10.15.

序号	坐标	光强
45	56.100	47.1
46	56.500	15.8
47	56.900	22.6
48	57.300	52.2
50	57.700	93.2
51	58.100	131.9
52	58.500	162.4
53	58.900	177.3
54	59.300	176.2
55	59.700	159.3
56	60.100	132.8
57	60.500	99.5
58	60.900	64.2
59	61.300	35.5
60	61.700	12.6
61	62.100	0.3
62	62.500	0.0

何锦源

3240104481

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2025.10.15

总审核人.

测一组 0级亮纹光强 3622.8
1级亮纹 182.7

$$\frac{182.7}{3622.8} = 5.04\%$$